Chapter 4
Professional Ethics

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Outline

- 4. Professional Ethics — 3 hours
  - 4.1. Profession
    - 4.1.1. Job and Occupation
    - 4.1.2. Characteristics of Profession
    - 4.1.3. Engineering and Computing as a Profession
  - 4.2. Professional Responsibilities and Rights
    - 4.2.1. Conflict of Interests and Whistleblowing
  - 4.3. Professional Code of Ethics
    - 4.3.1. Code of Ethics of Nepal Engineering Council
    - 4.3.2. Code of Ethics of IEEE and ACM
  - 4.4. Hacker Ethics and Netiquette

4.1 Profession : job and occupation

- Profess
  - To declare or admit openly
- Profession
  - An act of openly declaring or publicly claiming a belief, faith, or opinion
- Occupation:
  - a person's usual or principal work or business, especially as a means of earning a living
- Job
  - a piece of work, especially a specific task done as part of the routine of one's occupation or for an agreed price

So profession can be defined as :

- Vocation requiring advanced training and
  - generally recognized by universities and colleges as requiring special training of an advanced character leading to degrees distinct from the usual degrees in arts and sciences
  - requiring principally mental rather than manual or artistic labor and skill for its successful prosecution by reference to a common body of knowledge
  - recognizing the obligation of public service and of the public interest
  - having a code of ethics generally accepted as binding upon its members

Characteristics of a "Profession"

1. ESSENTIAL SOCIAL FUNCTION
   a. Recognition by society
   b. Presence of problems
   c. Grant of monopoly
   d. Image or prestige
   e. Recognition by employers
   f. Recognition by members
   g. Individuality
   h. Acceptance of responsibilities
2. SPECIALIZED BODY OF KNOWLEDGE
   a. Formal education
   b. Long and rigorous preparation
   c. Conceptual rather than practical
   d. Mental more than physical
   e. Unique language
3. REQUIREMENTS FOR ADMISSION
   a. Demonstration of technical competence
   b. Education
   c. Hurdle for monopoly rights
   d. Usually requires examination and experience
   e. Licensing often required

4. SELF-REGULATED
    a. Code of Ethics
    b. Character
    c. Ability to make difficult decisions
    d. Concern for “interests of society”
    e. Willingness to accept responsibility
    f. Procedures for discipline
5. ADAPTABILITY
    a. Continuing professional education
    b. Awareness of changing social conditions
    c. Significance of broad “liberal education”

Engineering and Computing as a Profession

Engineering is the profession in which knowledge of math and natural sciences gained by study, experience and practice is applied with judgment to economically utilize the materials and forces of nature for the benefit of mankind.

- **Characteristics and Responsibilities of Professional Engineers**
  - The engineer’s primary responsibility is to place safety of the public above all else.
  - Professional engineers possess education, knowledge, and skills that exceed those of the general public.
  - They must be abreast of discovers and technological changes by participation in professional meetings and continuing education.

- They must be willing to advance professional knowledge, ideals and practice.
- They have to share their knowledge with their peers.
- They must have a sense of responsibility and service to society and to their employers and clients.
- They must act honorably in their dealings with others.
- They must be willing to follow established codes of ethics for their profession. And to guard their professional integrity and ideals.

4.2 Professional Responsibilities and Rights

- In practice engineers' Responsibilities include much more than preventing and responding to accidents. In fact, during professional career of an engineer there are many responsibilities and rights.
- Responsibilities include both
  - Internal – responsibilities to employers and
  - External – responsibilities to the outside world

Internal responsibilities

- In today's competitive world, the success of any organization relies on its team-play. Working effectively as an engineer for a project requires the ethics of team-play. Team-play involves virtues of:
- 1. Collegiality`
- 2. Loyalty
- 3. Respect for authority and
- 4. Collective Bargaining.

Collegiality (team spirit)

- **Craig Ihara** defines collegiality as "a kind of connectedness grounded in *respect for professional expertise and in a commitment to the goals and values of profession".* It is the tendency to support and cooperate with the colleagues.
- Elements of collegiality
  - *Respect to the ideas and work of others:* This results in support and cooperation with one's colleagues. One gets back the support and cooperation in return, and this is mutually beneficial.
  - *Commitment to moral principles:* Commitment is towards moral decisions, actions, goals of the organization and values of the profession.
  - *Connectedness:* It means the shared commitment and mutual understanding. It ensures the absence of egoism and paves way for progress for both.
- Generally collegiality should be encouraged among engineers because
  - It is an influential value to promote the aims of professions. Therefore it strengthens an engineer's motivation to live up to professional standards.
  - It is more valuable as many individuals jointly working for the goodness of the public and society.
- Negative Aspects of Collegiality
  - It may be misused or distorted.
  - It may degenerate more groups of self-interest, rather than groups of shared devotion to the public good.
  - It may focus on corporate goal of maximizing profit at the expense of the public good.

Loyalty

- It is the quality of being true and faithful in one's support. It is more a function of attitudes, emotions and a sense of identity.
- *Two senses of Loyalty*
  1. Agency loyalty and
  2. Identification loyalty (Attitude loyalty)
  - AGENCY LOYALTY
    - ✓ It is an obligation to fulfill his/her contractual duties to the employer. The duties are specific actions one is assigned, and in general cooperating with others in the organization.
  - IDENTIFICATION LOYALTY
    - ✓ In contrast to agency loyalty, identification loyalty is much concerned with attitudes, emotions and a sense of personal identity as it does with actions.
    - ✓ Working together in falsification of records or serious harm to the public, does not merit loyalty

Respect for authority

- Authority can be defined as *the legal right to command action by others and to enforce compliance.*
- The authority fixes the personal responsibility and accountability uniquely on each person. This is necessary to ensure progress in action.
- INSTITUTIONAL AUTHORITY
- The characteristics features of institutional authority are that they allocate money and other resources and have liberty in execution.
- It is the right given to the employees to exercise power, to complete the task and force them to achieve their goals.

Collective bargaining

- International Labor Organization has defined collective bargaining as *"negotiation about working conditions and terms of employment between an employer and one or more representative employee's with a view to reaching agreement"*
- PROCESS OF COLLECIVE BARGAINING

1. Presenting the character of demands by the union on behalf of the constituent elements.
2. Negotiations at the bargaining table
3. Reaching an agreement

- **Arguments over Unions**
- There are two arguments in favor of and against unions.
- **In favor of unions**

❖ plays a vital role in achieving high salaries and improved standard of living of employees.

❖ Gives employees a greater sense of participation in organization decision making.

❖ Can act as counterforce to any radical political movements that exploit the employees.

- **Against unions**

❖ Shatter the economy of a country by placing distorting influences on efficient uses of labor.

❖ Remove negotiation between employers and employees.

EXTERNAL RESPONSIBILITIES

- The responsibilities to the outside world include:

1. Confidentiality
2. Conflict of Interest and
3. Occupational crimes.

CONFIDENTIALITY

- It is widely accepted that the engineers have an obligation to keep certain information of the employer/client secret or confidential. In the same way, engineers have an obligation to keep proprietary information of their employer/client confidential
- Confidential information is information deemed desirable to keep secret.
- Terms related to confidential information.
    - Privileged information – information available to an employee who is working on a special assignment.
    - Proprietary information – PROPERTY or OWNERSHIP - a new knowledge established within the organization that can be legally protected from use by others
    - Trade secrets – these are given limited legal protection against employee or contractor abuse.
    - Patents – legally protect specific products from being manufactured and sold by competitors.
- Types of information should be kept confidential are:
    - Information about the unreleased products.
    - Test results and data about the products
    - Design or formulas for products.
    - Data about technical processes.
    - Organization of plant facilities.
    - Business information

CONFLICTS OF INTEREST

- an individual has two or more desires that all interests cannot be satisfied given the circumstances.
- Professional conflicts of Interest are situations where professionals have an interest, if pursued, could keep them from meeting one of their obligations to their employers.
- Types of Conflicts of Interest
  - Actual Conflicts of Interest – loss of objectivity in decision- making and inability to faithfully discharge professional duties to employer.
  - *Potential conflicts of interest – may corrupt professional judgment in the future, if not in the present.*
  - *Apparent conflicts of interest – there are situations in which there is the appearance of a conflict of interest.*

- **Whistleblowing**
  - Whistleblowing is an act in which a person who informs on another or makes public disclosure of corruption or wrongdoing or any misconduct. The alleged misconduct may be classified in many ways; for example, a violation of a law, rule, regulation and/or a direct threat to public interest, such as fraud, health and safety violations, and corruption. Whistleblowers may make their allegations internally (for example, to other people within the accused organization) or externally (to regulators, law enforcement agencies, to the media or to groups concerned with the issues).
- Advantages of whistleblowing
  - **Public safety**
  - **Moral responsibility**
- Disadvantages of whistle blowing
  - **Retaliation**
  - **Conflicts of Interest**

Occupational Crime

- Occupational crimes are illegal acts made possible through one’s lawful employment.
- It is the secretive violation of laws regulating work activities.
- When committed by office workers or professionals, occupational crime is called ‘white collar crime’. These crimes are motivated by personal greed, corporate ambition, misguided company loyalty etc.
- These crimes impinge on various aspects such as professionalism, loyalty, conflicts of interest and confidentiality.
  - Examples of occupational crimes:
    - Price Fixing
    - Endangering Lives &
    - Industrial espionage(spying)

4.3 Professional Code of Ethics

- NEC
- IEEE
- ACM

Nepal Engineering Council

Discipline and Honesty: The Engineering service/profession must be conducted in a disciplined manner with honesty, not contravening professional dignity and wellbeing.

Politeness and Confidentiality: Engineering services for customers should be dealt with in a polite manner and professional information should remain confidential except with written or verbal consent of the customers concerned. This, however, is not deemed to be a restriction to provide such information to the concerned authority as per the existing laws.

Non-discrimination: No discrimination should be made against customers on the grounds of religion, sex, caste or any other things while applying professional knowledge and skills.

Professional Work: Individuals should only do professional work in their field or provide recommendation or suggestions only within the area of their study or obtained knowledge or skills. With regards to the works not falling within the subject of one's profession, such as works should be recommended to be done by an experts of the subject matter.

Deeds which may cause harm to the engineering profession: With the exception of salary, allowance, and benefits to be received for services provided, one shall not obtain improper financial gain of any kind of conduct improper activities of any kinds, which would impair the engineering profession.

Personal responsibility: All individuals will be personally responsible for all works performed in connection with his/her engineering profession.

State name ,designation and registration number: While signing the documents or descriptions such as the design , map , specification and estimates etc relating to the Engineering profession , the details should include, the name , designation and NEC registration No. and should be stated in a clear and comprehensive manner.

No publicity or advertisement must be made which cause unnecessary effects: In connection with the professional activities to be carried out, no publicity or advertisement shall be made so as to cause unnecessary effects upon the customers.

IEEE

1. to accept responsibility in making decisions consistent with the safety, health, and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding of technology; its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons and to not engage in acts of discrimination based on race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

4.4 Hacker Ethics

- Hacker: “A person with an enthusiasm for programming or using computers as an end in itself.” Or, “A person who uses his skill with computers to try to gain unauthorized access to computer files or networks.” – Oxford English Dictionary
- The term is misleading and has been used to mean many different things by computer professionals, cyber criminals and the media.
    - self-described hackers – enjoy experimenting with technology and writing code.
    - Media-labeled hackers (crackers) – break into systems, cause damage, and write malware.
    - Ethical hackers – former hackers or crackers who have joined the security industry to test network security and create security products and services.
- Black Hats – break into systems, develop and share vulnerabilities, exploits, malicious code, and attack tools.
- Grey Hats – are in hacker ‘no-man’s land’, may work as security professionals by day and ‘hack’ by night.
- White Hats – are part of the ‘security community’, help find security flaws, but share them with vendors so that products can be made safer.
- There is a fine line between the ‘hats’ and the distinction often becomes blurred. Often a matter of perspective.

Computer Ethics

- Thou shalt not use a computer to harm other people
- Thou shalt not interfere with other people's computer work
- Thou shalt not snoop around in other people's files
- Thou shalt not use a computer to steal
- Thou shalt not use a computer to bear false witness
- Thou shalt not use or copy software for which you have not paid
- Thou shalt not use other people's computer resources without authorization
- Thou shalt not appropriate other people's intellectual output
- Thou shalt think about the social consequences of the program you write
- Thou shalt use a computer in ways that show consideration and respect

Hackers responses

- That every individual should have the right to free speech in cyberspace.
- That every individual should be free of worry when pertaining to oppressive governments that control cyberspace.
- That democracy should exist in cyberspace to set a clear example as to how a functioning element of society can prosper with equal rights and free speech to all.
- That hacking is a tool that should and is used tot test the integrity of networks that hold and safeguard our valuable information.
- Those sovereign countries in the world community that do not respect democracy should be punished.
- That art, music, politics and crucial social elements of all world societies can be achieved on the computer and in cyberspace.

- That hacking, cracking, and phreaking are instruments that can achieve three crucial goals:
  - direct democracy in cyberspace
  - the belief that information should be free to all
  - The idea that one can test and know the dangers and exploits of systems that store the individual’s information.
- That cyber space should be a governing body in the world community, where people of all nations and cultures can express their ideas and beliefs has to how our world politics should be played.
- That there should be no governing social or political class or party in cyber space.
- That the current status of the internet is a clear example as to how many races, cultures, and peoples can communicate freely and without friction or conflict.